gheim-ch-pii-171k and gheim-ch-560m: An Open Swiss-Language PII Dataset and NER Model. Technical report.

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ABSTRACT

We release gheim-ch-pii-171k, a 171,336-chunk personally-identifiable information (PII) named-entity-recognition dataset covering the four official Swiss languages (Swiss German, Swiss French, Swiss Italian, Romansh) and English, and gheim-ch-560m, a 560M-parameter xlm-roberta-large fine-tune released alongside it. The dataset combines real Swiss text from court rulings, federal parliament records, Swiss-filtered web text and a Romansh corpus with an LLM-generated synthetic gap-fill layer that covers cells where the real-text corpus was too sparse for training and evaluation. As a generalization signal, gheim-ch-560m attains 0.89–0.96 person-NER F_1 on four widely-used external benchmarks (ai4privacy/openpii-1m, ZurichNLP swissner, Babelscape WikiNeural, CoNLL-2003), without ever seeing training data from any of them. On the in-distribution held-out test split it attains a strict-span F_1 of 0.916; among public detectors run zero-shot on the same Swiss test set, the strongest is openai/privacy-filter at 0.44. The dataset ships under CC BY 4.0; the model under Apache 2.0; both are available on the Hugging Face Hub. Reproducible eval scripts and checkpoints are on GitHub. This is an engineering report on a publicly-released artifact; no new method or architecture is proposed.

1 INTRODUCTION

The dominant pattern in production large-language-model (LLM) workflows is to call a hosted API: OpenAI, Anthropic, Google, Mistral. For Swiss-market deployments this raises a recurring compliance concern: every request that contains a customer name, a Swiss address, an IBAN, or an AHV/AVS social-security number leaves Swiss jurisdiction the moment it crosses the API boundary. The legal landscape for using third-party LLM services with personal data under Swiss law is analysed in detail by Rosenthal and Veraldi [19]; the practical mitigation in production systems is to anonymise sensitive content before the request and restore the originals on the response. The accuracy of the round-trip depends on the quality of the underlying named-entity recogniser.

Existing public PII detectors handle Swiss text poorly. The closest open generalist, openai/privacy-filter [15] (an OLMoE-based [13] 1.4B-parameter mixture-of-experts model with 50M active parameters), reaches a strict-span F_1 of 0.44 on our held-out Swiss test set despite multilingual training. Microsoft Presidio [12], configured with per-language NLP engines, reaches 0.43. Multilingual NER baselines such as Davlan's xlm-roberta-base-ner-hrl [1] reach 0.65 person-only F_1. None of the available systems fluently handle the four official Swiss languages together with the structured PII formats common in Swiss documents (CH-IBAN, AHV, VAT-CHE), and only one (SwissNER [24]) provides a Romansh evaluation slice.

This report documents three publicly-released artifacts:

(1) gheim-ch-pii-171k, a multilingual PII NER dataset of 171,336 chunks across Swiss German, Swiss French, Swiss Italian, Romansh and English, covering eight PII categories (account number, address, date, email, person, phone, URL, secret), encoded in a 33-class BIOES tag scheme byte-aligned with openai/privacy-filter.
(2) gheim-ch-560m, a 560M-parameter token-classification model fine-tuned from xlm-roberta-large [5] on the dataset, attaining a held-out test F_1 of 0.916 strict-span and 0.958 character-level.
(3) A reproducible evaluation harness that scores the released model on four established external benchmarks (ai4privacy/openpii-1m, ZurichNLP swissner, Babelscape WikiNeural [20], CoNLL-2003 [22]) and seven competing open detectors on the released test set, with replication checks against each baseline's published numbers on its native dataset.

1.1 Scope

This is a technical report on a deployed artifact, not a research contribution. We do not propose new model architectures, training methods, evaluation metrics, or annotation schemes. The model is xlm-roberta-large [5] fine-tuned with stock AdamW [10] and a small hyperparameter sweep; the label space is byte-identical to openai/privacy-filter [15]; the real-text corpora are taken as-is from the Apertus pretrain release [6] and downstream sources; the dataset construction pattern (LLM annotation followed by checksum-validated regex augmentation and template-based gap-fill) is by now standard practice. The contribution is the assembled artifact: a publicly-released Swiss-language PII dataset that includes Romansh, a strong out-of-the-box checkpoint for the same niche, and a reproducible evaluation that puts both into context against the open alternatives. The report documents what the dataset contains, how the model was trained, what its measured behaviour is on multiple test sets, and what its known limitations are, so downstream users can judge whether to deploy it.

The rest of the paper is structured as follows. Section 2 situates the work against prior art. Section 3 describes the dataset construction. Section 4 covers model selection and training. Section 5 reports the evaluation. Section 6 discusses limitations and Section 7 summarises. Appendix A collects the four full result tables (per-language × per-category breakdown, comparison to other detectors, cross-domain transfer, and a methodology-validation pass on each external baseline).

2 RELATED WORK

2.1 PII detection and de-identification

Microsoft Presidio [12] is a widely-used industrial open-source PII detector: a per-language spaCy NER engine combined with checksum-validated regex recognisers for structured formats (IBAN, credit card, SSN, etc.). Its built-in catalogue is large and the framework is easy to extend; its weakness on Swiss text is inherited from the per-language spaCy NER backbone (Section 5, Table 6).

The AI4Privacy family of corpora and fine-tuned checkpoints [2, 3] is a high-volume recent open-source effort. Their datasets are templated synthetic prose with embedded Faker-generated PII at the million-chunk scale across 12+ fine-grained categories and 17+ locale variants, and the published distilbert fine-tunes (e.g. Isotonic/distil-bert_finetuned_ai4privacy_v2) are widely-downloaded open PII detectors on the Hugging Face Hub. Section 5 confirms the published per-token accuracy numbers but shows a sharp drop on real Swiss text: when scored under strict-span F_1 on gheim-ch-pii-171k the fine-tunes reach 0.16. The cross-domain gap reflects template-vs-real domain shift more than label-noise: AI4Privacy's underlying templates were not designed with Swiss surface forms (CH-IBAN, AHV/AVS, +41 phones, German-language addresses) as targets.

openai/privacy-filter [15] is the strongest open zero-shot generalist on our test set. It is a token-classification head atop OL-MoE [13], a 1.4B-total-parameter mixture-of-experts encoder with 50M parameters active per token, and it shares our 33-class BIOES schema. Its `openai`-namespace release made it the default upstream from which our schema is byte-aligned, so a fine-tune started from this model can reuse the pretrained classifier-head weights.

2.2 Multilingual NER

XLM-RoBERTa [5] is a common starting point for multilingual token-classification and is the base of gheim-ch-560m. The Davlan family of fine-tunes [1] are trained on PER/ORG/LOC across many European languages and provide our generalist baseline. WikiNeural [20] provides large-scale silver-data multilingual NER training and evaluation across nine European languages and is one of our cross-domain external benchmarks. The CoNLL-2003 shared task [22] is a long-running English NER benchmark and is also part of our cross-domain evaluation. The BIOES tag scheme used throughout follows the design analysis of Ratinov and Roth [18].

2.3 Swiss-NLP and the Apertus stack

SwissBERT [23] is an encoder pretrained on Swiss text covering all four official languages, including Romansh. We considered SwissBERT as a base in the bake-off (Section 4, Table 3) and selected XLM-R-large by 0.8 pp validation F_1; SwissBERT is a competitive alternative when ~300M parameters is the deployment ceiling. Its companion test set, SwissNER [24], is the only public Swiss-news NER set that includes Romansh and is one of our cross-domain external benchmarks. The Romansh evaluation in particular relates closely to the WMT24++ benchmark extension of Vamvas et al. [25], which adds Rumantsch Grischun together with the five regional Romansh idioms (Sursilvan, Sutsilvan, Surmiran, Puter, Vallader) to the WMT machine-translation benchmark; the Apertus Romansh corpus, which underlies our `rm` chunks, is constructed in a similar idiom-aware way.

The real-text portion of our dataset is sourced from the Apertus pretrain corpora released alongside the Apertus Swiss-language LLM [6]. Apertus aggregates four Swiss-domain sources: the Open Legal Data Platform Entscheidsuche, which provides machine-readable Swiss court decisions across cantons in their original German, French, or Italian [4]; Curia Vista, the official business database of the Swiss Federal Assembly, which includes parliamentary motions, postulates, and transcripts in German, French, and Italian [21]; the Swiss-filtered subset of FineWeb-2 [17] (the multilingual web-crawl corpus that scales the FineWeb pipeline of Penedo et al. [16] to all languages and applies XLM-R-based quality filters); and a purpose-built Romansh corpus drawing on cantonal law texts, the Lia Rumantscha online dictionaries, the ZurichNLP bilingual corpus, and Romansh Wikipedia. These corpora are released under CC-BY-4.0-compatible licences and are the substrate for the real-Swiss-text portion of gheim-ch-pii-171k.

2.4 Quantisation for browser deployment

The released model ships an int8 ONNX export so the demo at gheim.ch runs entirely in-browser via WebGPU or WebAssembly. The quantisation procedure follows the post-training integer-arithmetic-only inference framework of Jacob et al. [9], applied through ONNX Runtime [11] via the `optimum` dynamic-quantizer with AVX-512 VNNI as the deployment target. The browser runtime is Transformers.js [8].

2.5 Industry tooling

transformers [26] is the training and inference runtime. seqeval [14] provides the strict-span scoring used both for our headline F_1 and for every comparison and replication entry in Section 5. spaCy [7] provides the per-language NER pipelines used in our comparison and inside Presidio. AdamW [10] is the optimiser. Faker_CH provides the Swiss-locale random data used in our synthetic gap-fill payload generation.

3 DATASET: GHEIM-CH-PII-171K

3.1 Source corpora and language coverage

Real-text content is drawn from the Apertus pretrain corpora [6]: the Swiss legal-decisions subset `entscheidsuche` curated from the Open Legal Data Platform of the same name [4], federal-parliament proceedings drawn from the `curia_vista` business database [21], the Swiss-filtered slice of FineWeb-2 [17], and the Romansh corpus released alongside Apertus, which compiles material from cantonal law texts, the Lia Rumantscha online dictionaries, the ZurichNLP bilingual corpus, and Romansh Wikipedia. Source documents are chunked at sentence boundaries to roughly 1,500 characters, then tagged with `facebook/fasttext-language-identification` at confidence ≥ 0.5. Chunks below threshold are dropped. The five-language distribution of the released dataset is shown in Table 1.

Table 1: Per-language chunk distribution.

Code	Variant	Chunks	%
de_ch	Swiss German (written std.)	48,564	28.3
fr_ch	Swiss French	48,097	28.1
it_ch	Swiss Italian	45,462	26.5
rm	Romansh	17,171	10.0
en	English (Swiss-context)	12,042	7.0
Total		171,336	100

Spoken-form Swiss German (GSW) is not represented: the corpus is written-standard German as used in Swiss legal and journalistic contexts. The fasttext language identifier labels GSW ("chuchichäschtli"-style orthography) as standard German, so incidental dialect content cannot be reliably separated.

3.2 Annotation schema

The label space follows the openai/privacy-filter taxonomy: eight PII categories listed in Table 2, each encoded in a BIOES tag scheme [18] (Begin, Inside, Outside, End, Single), giving 33 classes overall (one outside class plus four BIOES positions for each of the eight categories). The byte-identical label mapping was chosen for compatibility: it lets a fine-tune started from openai/privacy-filter's checkpoint reuse the pretrained classifier-head weights without remapping, and it lets downstream consumers swap the two checkpoints in place. The cost is that the upstream taxonomy's choices are inherited as-is, including ones we would have made differently. `account_number` lumps four structurally distinct formats (CH-IBAN, AHV/AVS, VAT-CHE, credit cards) into one cell, so the model cannot signal which kind it detected; `secret` is broad enough to catch any high-entropy token that looks key-shaped; there are no `organization` or `location` cells, which some downstream tasks would want. The `private_` prefix is historical; all eight cells are applied with a recall-oriented labelling policy and may flag publicly-listed institutional entities (e.g. a court switchboard phone number).

Table 2: The eight PII categories.

Category	Examples
`account_number`	CH-IBAN, AHV/AVS, VAT-CHE, credit cards
`private_address`	Bahnhofstrasse 1, 8001 Zürich
`private_date`	1990-01-02, 14. März 2026
`private_email`	alice@example.ch
`private_person`	Joel Barmettler, Dr. Schmidt
`private_phone`	+41 44 268 12 34
`private_url`	https://kanzlei.ch/...
`secret`	API keys, bearer tokens

3.3 Curation pipeline

Labelling. Real-text content is labelled by the Gemma 4 26B-A4B instruction-tuned model, run as a 4-bit AWQ quantisation (`cyankiwi/gemma-4-26B-A4B-it-AWQ-4bit`) via vLLM with structured-output JSON-schema decoding so that the output is directly parseable as span lists. The labeller was applied to the full Apertus pretrain corpora (approximately 4.5 million chunks across the five languages, around 14 hours of GPU wall time on a 2×RTX 4090 setup).

Regex augmentation. A separate pass re-scans the same Apertus corpora with a regex catalogue gated by checksum validators (ISO 13616 mod-97 for IBAN, EAN-13 mod-10 for AHV, ISO 7064 mod-11 for VAT-CHE, Luhn for credit cards). This is an independent scan rather than a post-filter on Gemma's output, so it can pick up structured PII in chunks that Gemma rejected or missed. Its output is merged with the Gemma annotations; on per-chunk overlap the regex annotator wins.

Balancing. The merged set of approximately 317k candidate chunks is passed through a four-phase balancer to limit document concentration, value memorisation and per-language class imbalance. The cap numbers were chosen from a prior pass of frequency analysis over the unbalanced output. Phase A caps each source document to 30 chunks. Phase B caps each (language, category, normalised value) tuple to 30 chunks, so a single recurring CH-IBAN or person name cannot dominate its cell. Phase C applies per-cell quotas tiered by language abundance: 8,000 chunks for each (de/fr/it) × (person, date, address, URL, phone) cell, 3,000 for de/fr/it × {email, account_number}, 2,000 for any Romansh cell, 1,500 for any English cell; `secret` is left uncapped because it is rare even in the source corpus. Phase D injects negative chunks (chunks with no detected PII) at 10 % of positives per language so the model trains on enough out-of-class context. The four phases together bring the 317k candidate chunks down to the 146,436 real-text chunks released in the dataset (approximately 46 % kept).

Synthetic gap-fill. A separate synthetic layer is appended to populate cells where real-Swiss-text coverage was insufficient. A slot-fill verifier pre-generates payload values (Faker_CH names, addresses and phones; checksum-validated CH-IBAN, AHV and VAT-CHE; common secret formats), then prompts Gemma to embed those exact values verbatim into natural multilingual prose (developer chat, customer-record notes, incident reports). The verifier accepts a chunk only if every payload value appears as an exact substring in the model output; rejection rates were a few percent per run. 24,900 of these synthetic chunks are placed in the training split (22,500), validation (1,200) and test (1,200), with deterministic seeded placement and disjoint chunk IDs across splits. The remaining 146,436 chunks are real-text. Every record carries a `synthetic` boolean field so consumers can filter to real-only or synthetic-only as required.

3.4 Splits and held-out integrity

Documents are partitioned at the source-document level before any chunk is taken, with stratification by (language × source), so no chunk in test or validation comes from a document with a chunk in train. Synthetic chunks are placed deterministically (random seed 17). Final split sizes are 139,641 / 15,834 / 15,861 chunks for train / validation / test.

3.5 Label-noise audit

The dataset's labels are machine-generated. No third-party human annotators were contracted, and no subset carries human gold labels. Every model F_1 reported in Section 5 is therefore a measure of agreement between the trained model and its machine-generated labelling source, not a measure of agreement with a human ground truth.

To put a number on the residual disagreement we ran a four-way re-labelling audit. A stratified 176-chunk sample was held back from training, re-annotated by four independent LLM subagents using a different prompt and a different seed for each, and the four labellings combined into a per-span majority vote. The released dataset's labels reach an F_1 of **0.66** against that majority vote. Inspection of the disagreements shows the gap is dominated by policy-interpretation differences (the most common case is whether the publicly-listed switchboard phone number of a court counts as personal information; the recall-oriented policy used for the dataset says yes, the modal subagent said no) rather than random annotator error or category confusion. The 0.66 audit is the best label-quality ceiling we can quote for this dataset, and the model's headline 0.916 strict F_1 should be read with that ceiling in mind: the model is matching the dataset's own labelling policy with high fidelity, which is not the same as matching a human's idea of what counts as PII.

4 MODEL: GHEIM-CH-560M

4.1 Architecture and base-model selection

The model is a token-classification head atop xlm-roberta-large [5] (24 layers, 1024 hidden, 16 heads, 560M parameters). XLM-R-large was selected following a controlled bake-off against ZurichNLP/swissbert [23] (270M dense), summarised in Table 3. Both bases received an identical 5×3 hyperparameter sweep over (learning rate, layer-wise LR decay) at one epoch, after which the winning configuration per base was trained for three full epochs and selected by best validation F_1. A third candidate, openai/privacy-filter, was deferred: the released model is required to ship as an ONNX export so that the in-browser demo at gheim.ch can run via transformers.js, and the Olmo-MoE architecture is not currently stable in optimum's [11] ONNX export path.

Table 3: Base-model bake-off.

Base model	Val F_1	Test F_1
ZurichNLP/swissbert (270M)	0.910	0.910
FacebookAI/xlm-roberta-large (560M)	**0.918**	**0.916**

In the (LR, LLRD) sweep, learning rate was the dominant factor: F_1 increased monotonically with LR over the tested range (5e-6 to 5e-5) for both bases. Layer-wise LR decay did not improve any cell at any tested value (1.0, 0.95, 0.9). The selected configuration is therefore LR = 5×10^{-5} without LLRD.

4.2 Training procedure

The training mix is the train split of `gheim-ch-pii-171k` (139,641 chunks) plus a small AI4Privacy English/email augmentation slice (approximately 14,000 chunks drawn from `ai4privacy/openpii-1m` [3]) added to shore up English-anchor coverage and CH-region email recall, for a total of 153,641 training chunks. The validation and test splits used for the headline metrics contain no AI4Privacy data. Hyperparameters are listed in Table 4.

Table 4: Training hyperparameters.

Hyperparameter	Value
Optimiser	AdamW [10]
Learning rate	5e-5
LR scheduler	Cosine, 5% warmup
Layer-wise LR decay	1.0 (off)
Effective batch size	128 (per-device 64 × 2 GPUs DDP)
Epochs	3; best at epoch 2.50
Max sequence length	512
Precision	bf16
Hardware	2 × NVIDIA RTX 4090
Wall time	≈ 52 min train + 4 min eval

The best checkpoint was selected by `eval_overall_f1` on the validation split. The held-out test split was scored exactly once on the selected checkpoint to produce the headline numbers in Section 5.

4.3 Deployment formats

The model ships in two formats. Server-side users load `model.safetensors` (fp32 PyTorch, 2.2 GB) via transformers [26]; in-browser users load `onnx/model_quantized.onnx` (int8 dynamic-quantised ONNX, 552 MB) via transformers.js [8] on WebGPU or WebAssembly. The int8 export reproduces 99.4% of the fp32 strict F_1 (0.909 vs 0.916, −0.7 pp). Per-category strict F_1 deltas range from +0.002 on `secret` to −0.053 on `account_number`, with the loss concentrated in the two categories that were already weakest under fp32 (`private_address` −0.021, `account_number` −0.053); the other six categories drop by less than 0.01.

5 EVALUATION

5.1 Methodology

Two metrics are reported throughout. **Strict-span F_1** is the standard NER literature metric: a predicted span counts as a true positive only if its (start, end, label) tuple matches a gold span exactly. Computed via seqeval [14] at the token level for HF and ONNX backends, and via char-level set match for span-emitting backends (spaCy, Presidio) that lack a token grid. **Char-level F_1** (label-aware) is computed over (character index, label) pairs: it asks "did the model mask the right characters with the right category?" and is invariant to entity-segmentation policy differences across systems.

The latter matters because cross-model PII evaluation has a structural problem: different annotation conventions split the same surface form differently. ai4privacy/openpii-1m splits a person mention into a GIVENNAME and a SURNAME span; gheim-ch-560m and CoNLL-2003 emit one combined PERSON span; address detectors fragment "Werdstrasse 36, 8004 Zürich" at the comma or do not. Strict-span F_1 penalises every boundary mismatch even when the underlying redaction-utility goal is met. Reporting both metrics lets the reader judge which is appropriate for their use case: strict for literature comparability, char for deployment.

5.2 In-distribution performance

The numbers in this subsection and the next come from a held-out test split drawn from the same Apertus source corpora as the training data, with strict document-level disjointness between splits but the same sampling distribution. They are an in-distribution upper bound rather than a generalization signal; the cross-domain numbers in Section 5.4 are the more meaningful evidence that the model transfers to data drawn from a different distribution.

On the held-out test split of gheim-ch-pii-171k (15,861 chunks), gheim-ch-560m attains a strict-span F_1 of 0.916 (precision 0.907, recall 0.926) and a char F_1 of 0.958. The validation/test gap is 0.002, consistent with no observable overfitting to the validation set used for checkpoint selection. The full per-language × per-category char F_1 breakdown is in Table 5 (Appendix A).

Structured categories with strong surface signals (`secret`, `email`, `phone`, `URL`) hold above 0.95 in every language; `account_number` accounts for most of the per-language variance. The Romansh `account_number` cell is the only one below 0.5: only 91 of its 117 gold spans across the test set are RM IBANs (the rest are synthetic-injected chunks), and the model under-recalls them in mixed-language contexts. Address weakness is more uniform across the four Swiss languages (0.83–0.92) and reflects boundary placement on multi-token addresses rather than category confusion. For production use both categories are intended to be paired with a deterministic regex front-end that applies checksum validation (see `packages/gheim-py/src/gheim/detectors/composite.py`).

5.3 Comparison to other systems

Seven other open PII / NER systems were scored on the same test split under the same harness; the full result table is in Table 6 (Appendix A). PER-only models (Davlan, dslim, spaCy) emit only person mentions and no other PII categories, so their overall 8-category F_1 is reported as n/a; the meaningful number for those is the `private_person` cell.

Of the seven public detectors we measured on Swiss text in this configuration, gheim-ch-560m is the strongest on every cell that is directly comparable. The comparison is asymmetric: gheim-ch-560m is in-distribution on this test set (the training data was drawn from the same corpora) and the seven contestants are all run zero-shot, so the absolute gap is partly a domain-fit effect, not only a model-quality difference. With that caveat noted, the two metrics tell the same ranking story: under strict F_1 the next-strongest entry is openai/privacy-filter at 0.44, under char F_1 at 0.61. The gap between the two metrics is largest for systems whose schema differs most from ours: Isotonic/distilbert_finetuned_ai4privacy_v2 gains +35 pp under char F_1 because its fine-grained schema (FIRSTNAME, LASTNAME, MIDDLENAME, PREFIX, ...) collapses under our 8-category remap and produces highly fragmented spans; openai/privacy-filter gains +17 pp because the Olmo tokeniser emits subword boundaries that do not align with the gold span ends.

Among the contestants, gheim-ch-560m is also the only entry with native Romansh support; spaCy has no Romansh or English pipelines in this configuration, and dslim/bert-base-NER is English-only. Section 5.4 reports the same model on each contestant's training distribution for the reverse comparison.

5.4 Cross-domain evaluation

The released model was scored on four established external benchmarks (ai4privacy/openpii-1m, Babelscape WikiNeural, ZurichNLP swissner, CoNLL-2003); the full result table is in Table 7 (Appendix A). For pure-NER datasets (swissner, CoNLL-2003, WikiNeural) only PER maps to a gheim category, so the meaningful number is the PER cell. To anchor the gheim cells, the table reports alongside each row the F_1 of a model originally trained on that benchmark's distribution (reproduced through our harness).

Row by row, gheim-ch-560m matches or exceeds three of the four specialists despite being run zero-shot and without ever seeing training data from those benchmarks; only on CoNLL-2003 does the English-only specialist remain ahead, by 7 pp char and 7 pp strict. The notable strict-vs.-char gap on openpii-1m (0.776 strict, 0.920 char) is entirely a schema artefact: openpii-1m splits each person into separate GIVENNAME and SURNAME gold spans, which fragment into two adjacent `private_person` spans under our remap even though every PII character is correctly masked. On Swiss news (swissner, the only Swiss-news NER set that includes Romansh), per-language person-character F_1 is 0.864 (de), 0.870 (fr), 0.835 (it), 0.795 (rm).

A methodology check that replicates each baseline on its own training-distribution dataset and compares the reproduction to its published number is reported in Appendix A (Table 8). Three of four reproductions match within ±0.03; the fourth (Isotonic) is a metric mismatch (per-token accuracy versus strict-span F_1) rather than a harness bug.

6 LIMITATIONS

- **Recall-oriented labelling policy.** Public Swiss documents contain extensive personally-identifiable information that is already public (court rulings, parliament records, public officials). The dataset labels these as PII and the model learns to flag them. Applications that need to distinguish public from private entities should layer a downstream public-vs-private classifier.
- **`private_address` test F_1 is 0.78.** Boundary placement on multi-token addresses is the dominant error mode.
- **`account_number` test F_1 is 0.67.** Production deployments should pair the model with a regex front-end with checksum validation (IBAN, AHV, VAT-CHE, Luhn).
- **Romansh test F_1 is 0.85**, the weakest of the five languages. The Romansh training material is dominated by a

single literary/journalistic register (Rumantsch Grischun), so dialectal or technical RM text is unmeasured.

- **Swiss German dialect (GSW) is not measured.** The fasttext detector used in data preparation labels GSW as standard German.
- **Annotations are machine-generated.** The dataset was not human-annotated; the audit number in Section 3.5 (F_1 0.66 against a four-way subagent re-labelling) is the closest thing to a labelling-policy ceiling we can quote.
- **Re-identification is not in scope.** The model is intended for redaction. It is not designed to identify or link entities to real persons and does not return entity-linked identifiers.

Inference-time suppression and the calibrated decoder. A failure mode visible only on deploy probes, not in the test-set metrics: on a small fraction of inputs the raw model emits zero non-O tokens for an entire chunk, even when the chunk contains clear PII. The pattern reproduces consistently when a low-confidence person mention is immediately followed by a structurally distinctive token (e.g. a CH-IBAN), which appears to nudge the per-token argmax over the entire span toward the dominant `O` class. The released SDKs ship a thin wrapper that subtracts a small constant bias from the `O`-class logit before argmax (`CalibratedDetector`, default bias 0.5). On the failing probe set this recovers approximately 95% of the suppression cases; on the in-distribution test split it costs approximately 0.4 pp strict F_1. The wrapper is the default detector in both the Python and JavaScript SDKs. The published checkpoint exhibits a known inference-time failure mode that the test-set numbers do not expose; the wrapper trades a small amount of in-distribution F_1 for a noticeable improvement on the failure mode without requiring retraining.

7 SUMMARY

This report described gheim-ch-pii-171k and gheim-ch-560m, a Swiss-tuned PII NER dataset and model targeting LLM-API anonymisation round-tripping, a deployment context that existing open detectors handle poorly on Swiss text. On the in-distribution test split the model attains a strict-span F_1 of 0.916; among the seven public detectors we measured zero-shot on the same set the strongest is at 0.44. As an out-of-distribution generalization signal the model attains person-NER F_1 between 0.89 and 0.96 across four external benchmarks (ai4privacy/openpii-1m, swissner, WikiNeural, CoNLL-2003) without ever seeing training data from any of them. The dataset is released under CC BY 4.0; the model under Apache 2.0; both are on the Hugging Face Hub. The full evaluation harness, the per-system JSON outputs, and the reproducible build of the comparison tables are at https://github.com/joelbarmettlerUZH/gheim.

Known follow-on work. The dataset has no coverage of spoken Swiss German (the fasttext language identifier labels GSW as standard German) and only one Romansh register (Rumantsch Grischun); both gaps would require additional source corpora rather than pipeline changes. The model's released checkpoint is 560M parameters, larger than is convenient for edge or browser deployment; a smaller distilled variant would be useful but is not part of this release. The dataset's recall-oriented labelling policy flags publicly-listed institutional entities (e.g. a court switchboard phone) as PII; applications that require stricter precision would benefit from a downstream public-vs-private entity classifier layered on top.

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A DETAILED EVALUATION RESULTS

This appendix contains the four full result tables referenced from Section 5: the in-distribution per-language × per-category breakdown for gheim-ch-560m; the comparison to seven other public detectors on the same test split; the cross-domain transfer numbers on four external benchmarks; and a methodology check that re-runs each external baseline on its own training distribution. The prose in Section 5 summarises what these tables show; the tables themselves are kept out of the main narrative so each can be inspected at full width.

A.1 In-distribution per-language × per-category

Table 5 gives the per-language × per-category char-level F_1 on the held-out test split of gheim-ch-pii-171k. Structured categories with strong surface signals (`secret`, `email`, phone, URL) hold above 0.95 in every language; `account_number` carries most of the per-language variance, with the Romansh cell as the only one below 0.5.

A.2 Comparison to other detectors

Table 6 reports seven other open PII / NER systems on the gheim-ch-pii-171k test split under the same harness. Both the overall 8-category metric and the `private_person` cell are shown for each system; PER-only systems have n/a for the overall metric since they cannot emit the other seven categories. The comparison is in-distribution for gheim-ch-560m and zero-shot for the seven contestants; see Section 5.4 for the reverse direction.

A.3 Cross-domain transfer

Table 7 reports gheim-ch-560m PER F_1 on four external benchmarks the model has never seen. To anchor each row, the same column reports a model trained on (or claimed-state-of-the-art on) that benchmark's native distribution, reproduced through our harness. The footnotes give the specific metric / slice each native number is computed on.

†Isotonic's strict-span F_1 on its own training distribution; per-token accuracy across all categories (the metric Isotonic itself reports) is 0.969 in our reproduction versus 0.955 claimed. ‡Davlan-XLMR's overall multilingual avg from our reproduction (claimed ~0.84). §spaCy de_core's overall F_1 on the German slice of swissner (spaCy ships per-language pipelines and has no single multilingual model that covers the whole set).

Table 5: Per-language × per-category char-level F_1 on the held-out test split of gheim-ch-pii-171k under gheim-ch-560m. The right-most column gives the per-category average (over languages, gold-weighted); the bottom row gives the per-language average (over categories).

Category	de_ch	fr_ch	it_ch	rm	en	Avg.
`account_number`	0.932	0.717	0.660	0.169	0.994	0.765
`private_address`	0.890	0.870	0.915	0.825	0.973	0.889
`private_date`	0.943	0.934	0.952	0.888	0.909	0.939
`private_email`	0.988	0.994	0.999	0.992	0.999	0.994
`private_person`	0.938	0.948	0.962	0.897	0.951	0.944
`private_phone`	0.989	0.985	0.993	0.995	0.997	0.990
`private_url`	0.992	0.994	0.994	0.958	0.980	0.992
`secret`	0.999	1.000	0.999	1.000	1.000	1.000
Avg.	0.954	0.953	0.972	0.923	0.981	0.958

Table 6: Other models on the gheim-ch-pii-171k test split. Strict and char F_1 for both the overall 8-category metric and the `private_person` cell.

Model	Strict F1	Char F1	PER strict	PER char
joelbarmettler/gheim-ch-560m	**0.916**	**0.958**	**0.915**	**0.944**
openai/privacy-filter (1.4B MoE, zero-shot)	0.443	0.610	0.406	0.561
Microsoft Presidio (multi-lang config)	0.434	0.562	0.442	0.521
Davlan/xlm-roberta-base-ner-hrl	n/a	n/a	0.645	0.728
Davlan/distilbert-base-multilingual-cased-ner-hrl	n/a	n/a	0.619	0.771
spaCy `de/fr/it_core_news_lg`	n/a	n/a	0.505	0.621
dslim/bert-base-NER (English slice)	n/a	n/a	0.512	0.770
Isotonic/distilbert_finetuned_ai4privacy_v2	0.157	0.508	0.264	0.507

Table 7: gheim-ch-560m PER F_1 on four external benchmarks, anchored against a specialist model trained or claimed-state-of-the-art on each benchmark's native distribution. gheim-ch-560m is run zero-shot; native baselines are reproduced through the same harness.

External benchmark	*n*	gheim PER strict	gheim PER char	Native baseline	Native F_1
ai4privacy/pii-masking-openpii-1m	8,000	0.776	0.920	Isotonic/distilbert_ai4privacy	0.78[†]
Babelscape WikiNeural	8,000	0.919	0.960	Davlan/xlm-r-base-ner-hrl	0.813[‡]
ZurichNLP swissner	800	0.903	0.946	spaCy `de_core_news_lg`	0.841[§]
CoNLL-2003	3,453	0.887	0.936	dslim/bert-base-NER	0.957

Table 8: Each baseline scored on its own training-distribution dataset, against published numbers.

Replication target	Claimed	Repro	Δ
dslim/bert-base-NER on CoNLL-2003	0.910	0.913	+0.003
dslim PER on CoNLL-2003	∼0.96	0.957	−0.003
Davlan-XLMR on WikiNeural avg	∼0.84	0.813	−0.027
spaCy de_core on swissner_de	∼0.83	0.841	+0.011
Isotonic distilbert on AI4Privacy	0.955	0.969[†]	+0.014

A.4 Methodology validation: replicating each baseline on its own training distribution

To check that the harness in Section 5 is sound rather than an unfair implementation, we replicated each external system on the dataset on which it was trained and compared our reproduction to the system's own published F_1 (Table 8). Three of four reproductions match within ±0.03; the fourth (Isotonic) is a metric mismatch discussed below.

[†]Isotonic's claimed F_1 of 0.955 initially reproduced as 0.78 strict-span F_1 on the same data, an apparent 17-pp gap. The difference traces to the metric: Isotonic's reported number is *per-token classification accuracy* (which includes the dominant `O` class and is much higher than per-span F_1). Reproducing Isotonic's metric exactly (per-token accuracy including `O`) yields 0.969, matching their 0.955 within sampling noise. Our strict-span F_1 of 0.78 on the same data is the correct NER-literature comparison.